Inverse Functions & Parametrics
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 If $f(x) = \frac{3+e^{2x}}{5}$, find an expression for $f^{-1}(x)$. 2

$y = \frac{3+e^{2x}}{5}$

Inv: $x = \frac{3+e^{2x}}{5}$

$5x = 3+e^{2x}$

$5x - 3 = e^{2x}$

$2x = \ln(5x-3)$

$x = \frac{1}{2}\ln(5x-3)$

2 Given the function $f(x) = \sqrt{2x+1}$ and that $f^{-1}(x)$ is the inverse function of $f(x)$, find $f^{-1}(5)$. 2

$5 = \sqrt{2x+1}$

Squaring both sides,

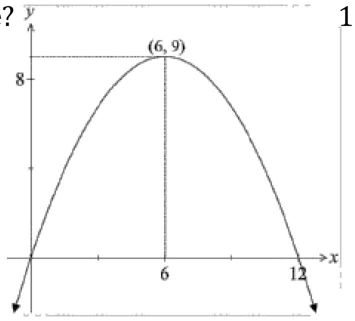
$25 = 2x+1$

$2x = 24$

$x = 12$

$\therefore f^{-1}(5) = 12$

3 Which of the parametric equations below represents the parabola above? 1



A. $x = 12t, y = 9t^2$ B. $x = 12t, y = 9 - t^2$

C. $x = 6 - 2t, y = 9 - t^2$ D. $x = 2t - 6, y = 9t - t^2$

A If $x = 12$, $t = 1$ $\therefore y = 9$ ✗

B If $x = 6$, $t = \frac{1}{2}$ $\therefore y = 8\frac{3}{4}$ ✗

C If $x = 0$, $t = 3$ $\therefore y = 0$ OK

If $x = 6$, $t = 0$ $\therefore y = 9$ OK

If $x = 12$, $t = -3$ $\therefore y = 0$ OK

D If $x = 0$, $t = 3$ $\therefore y = 18$ ✗

4 Find the derivative of the inverse function $f^{-1}(x) = 2x(x+6)^6$. 3

Inverse function is $x = 2y(y+6)^6$

$\frac{dx}{dy} = u'v + v'u$

where $u = 2y \quad u' = 2$

$v = (y+6)^6 \quad v' = 6(y+6)^5$

$$\frac{dx}{dy} = 2 \cdot (y+6)^6 + 6(y+6)^5 \cdot 2y$$

$$= 2(y+6)^6 + 12y(y+6)^5$$

$$= 2(y+6)^5(y+6+6y)$$

$$= 2(y+6)^5(7y+6)$$

$$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{2(y+6)^5(7y+6)}$$

5 A curve C has parametric equations $x = cos^2 t$ and $y = 4sin^2 t$ for $t \in R$. What is the Cartesian equation of C? 2

$x = \cos^2 t$ ...①

$y = 4\sin^2 t$ ...② $\rightarrow \frac{y}{4} = \sin^2 t$ ...③

① + ③ $x + \frac{y}{4} = \cos^2 t + \sin^2 t$

$x + \frac{y}{4} = 1$

$4x + y = 4$

But $0 \leq \cos^2 t \leq 1$

$\therefore 0 \leq x \leq 1$

$\therefore y = 4 - 4x$ for $0 \leq x \leq 1$

6 Consider the function $f(x) = x^2 - 4x + 6$.

a. Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function. 1

$f(x) = x^2 - 4x + 6$ is a parabola
Except for the turning point $(2,2)$
for each value of $f(x)$ in the range
there are two x-values. OR
ie fails the horizontal line test

b. Given that the domain of $f(x)$ is restricted to $x \leq 2$, find an expression for $f^{-1}(x)$. 2

Note $x \leq 2$
interchange x and y
$x = y^2 - 4y + 6$
$x - 6 = y^2 - 4y$
$x - 6 + 4 = y^2 - 4y + 4$
$x - 2 = (y-2)^2$
$\therefore y - 2 = \pm\sqrt{x-2}$

(Note discard $\sqrt{x-2}$ as $y \leq 2$)
$\therefore y = -\sqrt{x-2} + 2$
$f^{-1}(x) = -\sqrt{x-2} + 2$

c. Given the restriction in part (a) (ii), state the domain and range of $f^{-1}(x)$. 2

Domain: $x \geq 2$ as $x - 2 \geq 0$
Range: $y \leq 2$ as $-\sqrt{x-2} \leq 0$

d. The curve $y = f(x)$ with its restricted domain and the curve $y = f^{-1}(x)$ intersect at the point P. Find the coordinates of P. 2

$y = f(x)$ and $y = f^{-1}(x)$ have
a common intersection with the line $y = x$
$\therefore x^2 - 4x + 6 = x$
$x^2 - 5x + 6 = 0$
$(x-3)(x-2) = 0$
$\therefore x = 2, 3$
When $x = 2$ $y = 2$ and ~~so~~
$(2,2)$ lies on the line $y = x$
When $x = 3$ $y = 1$ does not lie
of the line $y = x$
$\therefore$ co-ordinates of P are $(2,2)$

7 The table shows selected values of a one-to-one differentiable function $g(x)$ and its derivative $g'(x)$. Let $f(x)$ be a function such that $f(x) = g^{-1}(x)$. Find the value of $f'(-1)$. 3

x	-1	0
$g(x)$	-5	-1
$g'(x)$	3	$\frac{1}{2}$

From the table $f(x) = g^{-1}(x)$
so $f(-1) = g^{-1}(-1) = 0$
$f'(-1) = \dfrac{1}{g'(f(-1))} = \dfrac{1}{g'(0)} = \dfrac{1}{\frac{1}{2}} = 2$

8 Given the parametric equations $x = sin^{-t}\, t + 1$ and $y = \frac{1}{\sqrt{1-t^2}}$. What is the Cartesian equation? 2

$x - 1 = \sin^{-1} t.$

$\sin(x-1) = t.$

$y = \dfrac{1}{\sqrt{1-(\sin(x-1))^2}} = \dfrac{1}{\cos(x-1)}$

9 An equation can be expressed in the parametric form 2

$$x = 2cos\theta - 1$$

$$y = 2 + 2sin\theta$$

Express the equation in Cartesian form.

$x = 2\cos\theta - 1 \Rightarrow \cos\theta = \dfrac{x+1}{2}$

$y = 2 + 2\sin\theta$

$\Rightarrow \sin\theta = \dfrac{y-2}{2}$

Using $\cos^2\theta + \sin^2\theta = 1$

$\left(\dfrac{x+1}{2}\right)^2 + \left(\dfrac{y-2}{2}\right)^2 = 1$

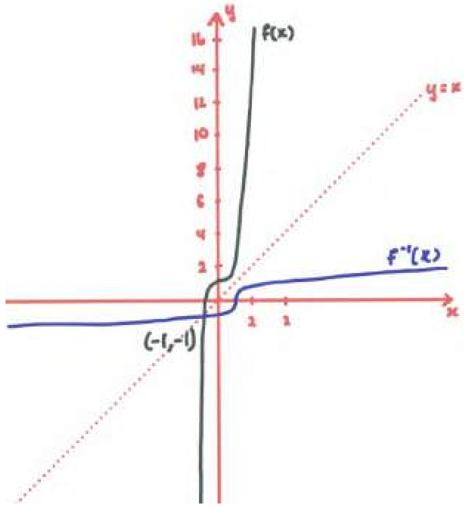
$\dfrac{(x+1)^2}{4} + \dfrac{(y-2)^2}{4} = 1$

$(x+1)^2 + (y-2)^2 = 4$

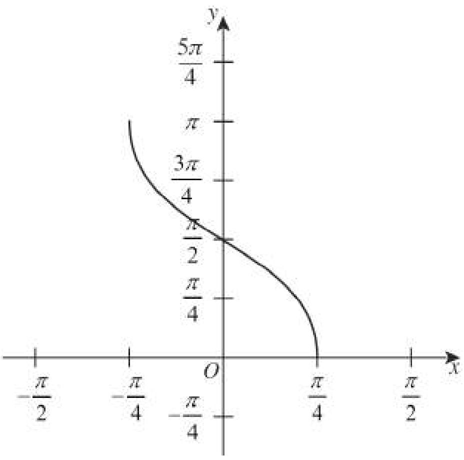
10 Given that $f(x) = e^x - 1$, and $y = f^{-1}(x)$, find an expression for $\frac{dy}{dx}$. $\dfrac{1}{x+1}$ 2

11 Sketch the inverse $y = f^{-1}(x)$ given the function 2

$f(x) = 2x^3 + 1.$



12 The graph of an inverse trigonometric function is shown. 2



Domain:

$-1 \le \dfrac{4x}{\pi} \le 1$

$-\pi \le 4x \le \pi$

$-\dfrac{\pi}{4} \le x \le \dfrac{\pi}{4}$

Range:

$0 \le y \le \pi$

$y = \cos^{-1}\left(\dfrac{4x}{\pi}\right)$

Inverse Functions & Parametrics
Topic Test 2 Solution

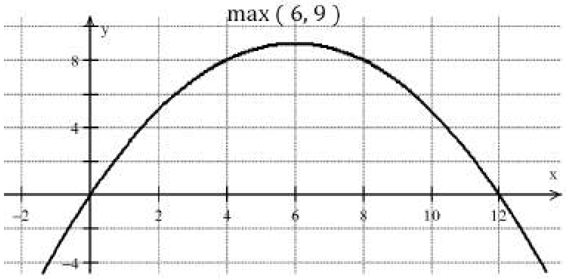
Total Marks: 30

Time: 50 minutes

1 A function is defined by $f(x) = (x+3)^2 + 1$ for $x \leq -3$. What is the inverse function of $f(x)$? 1

$$f^{-1}(x) = -3 - \sqrt{x-1}$$

2 Which of the parametric equations below represents the parabola above? 1



A. $x = 12t, y = 9t$

B. $x = 12t, y = 9 - t^2$

C. $x = 6 - 2t, y = 9 - t^2$

D. $x = 2t - 6, y = 9t - t^2$

test (0,0) & (12,0) in each one

3 Find the Cartesian equation of the curve with the parametric equations: 3

$$x = 4cos\theta - sin\theta$$

$$y = 5cos\theta + sin\theta$$

$x = 4\cos\theta - \sin\theta$ (1)
$y = 5\cos\theta + \sin\theta$ (2)

(1) + (2) : $\boxed{x + y = 9\cos\theta}$ (a) ✓

(1) × 5 : $5x = 20\cos\theta - 5\sin\theta$ (3)

(2) × 4 : $4y = 20\cos\theta + 4\sin\theta$ (4)

(4) − (3) : $\boxed{4y - 5x = 9\sin\theta}$ (b)

Adding square (a) and square (b)

$x^2 + 2xy + y^2 + 16y^2 - 40xy + 25x^2 = 81$ ✓

OR $26x^2 + 17y^2 - 38xy - 81 = 0$

4 Consider the function $f(x) = x + e^{5x} + 2$ and its inverse function $g(x)$. Use the derivative of $f(g(x))$ to determine the gradient of the tangent to the curve $y = g(x)$, at the point where $x = 3$. 2

$f(x) = x + e^{5x} + 2$

$f^{-1}(x) = g(x)$ OR $f[g(x)] = x$

For $f(0,3) \Rightarrow g(3,0)$

Also $f'(x) = 1 + 5e^{5x}$

$f'[g(x)] = f'[g(x)] \cdot g'(x) = 1$

$f'[g(3)] \cdot g'(3) = 1$

$f'(0) \cdot g'(3) = 1$

But $f'(0) = 1 + 5e^{5\times 0} = 6$

$g'(3) = \frac{1}{f'(0)} = \frac{1}{6}$

5 Which of the following pairs of parametric equations represents a circle that passes through the origin? 1

A. $x = 3 + 3cos\theta$, $y = 4 + 3sin\theta$ B. $x = 3 + 4cos\theta$, $y = 4 + 4sin\theta$

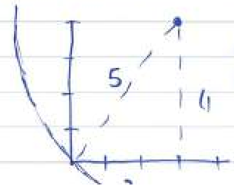
C. $x = 3 + 5cos\theta$, $y = 4 + 5sin\theta$ D. $x = 3 + 7cos\theta$, $y = 4 + 7sin\theta$

$(x-3)^2 + (y-4)^2 = 25\cos^2\theta + 25\sin^2\theta$

$(x-3)^2 + (y-4)^2 = 25(\cos^2\theta + \sin^2\theta)$

$(x-3)^2 + (y-4)^2 = 25.$

$C(3,4)$ $r = 5.$



$h^2 = 4^2 + 3^2$

$h = 5.$

(C)

6 Consider the function $f(x) = 2 - log_e x$.

a. Find the equation of the inverse function $f^{-1}(x)$. 1

Let $y = 2 - \log_e x$

Then for inverse:

$x = 2 - \log_e y$

$\log_e y = 2 - x$

$y = e^{2-x}$

b. Explain why the coordinate of the point of intersection P of the graphs $y = f(x)$ and $y = f^{-1}(x)$ satisfies the equation $e^{2-x} - x = 0$. 3

To find the point of intersection P, solve simultaneously:

$y = 2 - \log_e x$ —— ① Rearranging ①: $2 = y + \log_e x$

$y = e^{2-x}$ —— ② Rearranging ②: $2 = x + \log_e y$

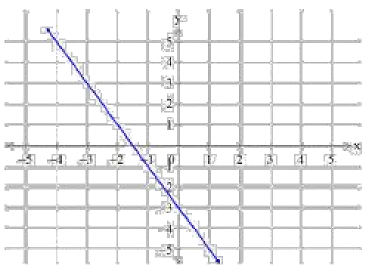
Equating $\Rightarrow$ $y + \log_e x = x + \log_e y$

which has $x = y$ as a solution.

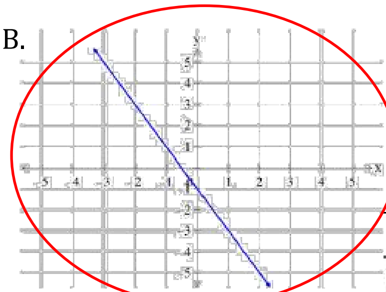
$\therefore$ P must satisfy $e^{2-x} = x$ or $e^{2-x} - x = 0$

7 What is the graph of thse function which has parametric equation $x = -\frac{p}{2} - 1, y = p + 1$? 1

A.



B.

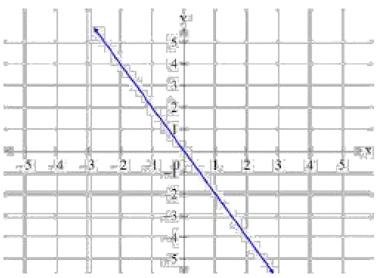


eg when $p = 2$, $x = -2$, $y = 3$

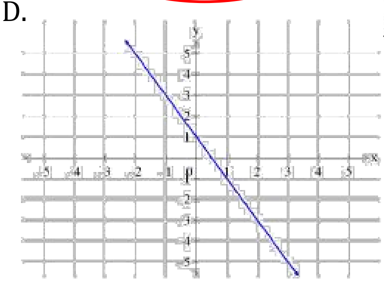
$p = 0$, $x = -1$, $y = 1$

Find the graph on which points $(-2,3)$ and $(-1,1)$ lie.

C.



D.



8 Consider the curve $f(x) = x^2 - 4x + 5$.

a. Find the largest possible positive domain for which $f(x)$ has an inverse function $f^{-1}(x)$. 1

$f(x) = x^2 - 4x + 5 \Rightarrow$ Axis of symmetry $= \frac{4}{2}$

$x = 2$

$\therefore$ largest positive domain : $x \geq 2$

b. Find the point(s) of intersection of $y = f(x)$ and $y = f^{-1}(x)$ in the domain determined in part (i). 2

Points of intersection are on the line $y = x$

i.e. when $x = x^2 - 4x + 5$

$x^2 - 5x + 5 = 0$

$x = \frac{5 \pm \sqrt{25 - 20}}{2} = \frac{5 \pm \sqrt{5}}{2}$

In the restricted domain point of intersection is

$\left(\frac{5+\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$

c. State the domain of $y = f^{-1}(x)$. 1

For $f(x)$, when $x = 2$, $y = 1 \Rightarrow$ range is $y \geq 1$

$\therefore$ Domain of inverse function is $x \geq 1$.

d. What is the equation of $y = f^{-1}(x)$? 2

$f^{-1}(x)$: $x = y^2 - 4y + 5$

$x = y^2 - 4y + 4 + 1$

$x - 1 = (y-2)^2$

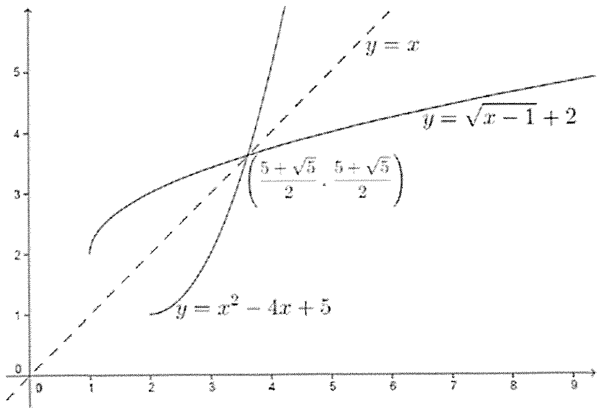
$y - 2 = \pm\sqrt{x-1}$

$y = 2 \pm \sqrt{x-1}$

Since range of inverse is $y \geq 2$

$y = 2 + \sqrt{x-1}$

e. For the restricted domain, sketch the function and the inverse function on the same number plane. Clearly label the graphs and show the main features. 3



9 A circle is represented by the parametric equations $x + 2 + 4cos\theta, y = 3 + 4sin\theta$. What are the centre and radius of the circle? 1

Centre $(2,3)$, radius 4.

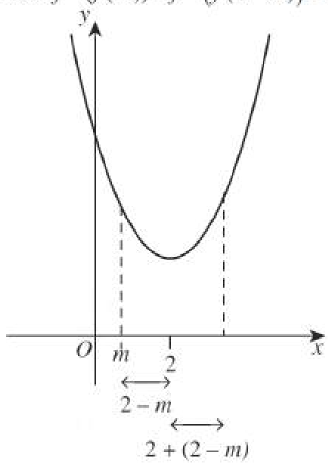
10 The function $f(x) = x^2 - 4x + 7$ has an inverse function $f^{-1}(x)$ in the domain $x \geq 2$. What is the value of $f^{-1}(f(m))$, where m is a real number NOT in the domain $x \geq 2$? 2

A. m B. $2 - m$

C. $2 + m$ D. $4 - m$

$f(x)$ maps m outside the domain to $2 + 2 - m = 4 - m$, which is inside the domain.

Now $f^{-1}(f(m)) = f^{-1}(f(4-m)) = 4 - m$.



9 A curve is described by the parametric equations: $x = \frac{t}{2}$ and $y = 3t^2$ 1

What is the cartesian equation of the curve?

$x = \frac{t}{2}$ ①

$y = 3t^2$ ②

From ① $t = 2x$ ③

sub ③ into ②

$y = 3 \times (2x)^2 = 12x^2$

10 What is the equation of $y = f^{-1}(x)$ if $y = x^2 - 2x - 2$ where $x \geq 1$? 1

A. $y = 1 \pm \sqrt{x+3}$ B. $y = 1 + \sqrt{x-3}$

C. $y = 1 - \sqrt{x+3}$ D. $y = 1 + \sqrt{x+3}$

$y = x^2 - 2x - 2$

$x = y^2 - 2y - 2$

$x = y^2 - 2y + 1 - 1 - 2$

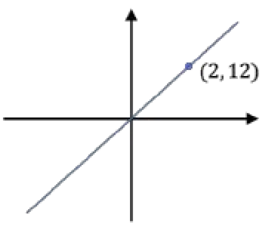
$= (y-1)^2 - 3$

$(y-1)^2 = x + 3$ $\therefore P(2,-2), P'(-2,2)$

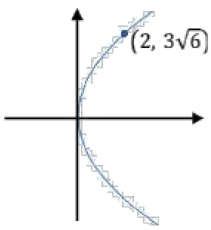
$y = 1 \pm \sqrt{x+3}$ $\therefore f^{-1}(x) = 1 + \sqrt{x+3}$ D

11 A function is defined by the parametric equations, $x = 3t^2$ and $y = 9t, t > 0$. Which of the following sketches represents this function? 1

A.



B.

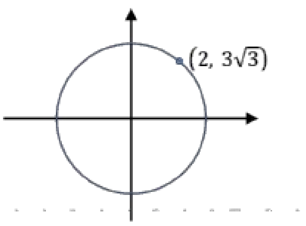


for $x=2$, $2 = 3t^2$

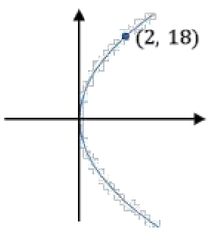
$t = \sqrt{\frac{2}{3}}$

$y = 9\sqrt{\frac{2}{3}} = \frac{9\sqrt{6}}{3} = 3\sqrt{2}$

C.



D.



12 A function is described by the parametric equations $x = t + 1$ and $y = t^2 - 1$. Find the cartesian equation of this function. 2

$x = t + 1$ ① ; $y = t^2 - 1$ ②

$t = x - 1$

substitute $t = x - 1$ into ②

$y = (x-1)^2 - 1 = x^2 - 2x + 1 - 1$

$y = x^2 - 2x$